MLOps PROJECT / INDIA / DEPLOYED 27 OCT 2026

DELHI AQI FORECASTING

Tomorrow's air quality.

A reproducible forecast, monitored after deployment.

PRACTICAL PURPOSE

Predict next-day city AQI to support awareness and planning around pollution trends.

REAL DATA / EASY ACCESS

3,143 training observations

Training: 2015-2023 / fresh CPCB daily bulletins
Original MAE: forest 31.09 / persistence 31.61

01 ML PIPELINE + CI/CD

Poster criterion 1 / CO4

01 CPCB CSV — Version + validate
→ **02 Lag features** — Seven prior AQI days
→ **03 Train + compare** — Persistence vs forest
→ **04 MLflow tracking** — Params + run metrics
→ **05 Quality gate** — Chronological holdout
→ **06 GitHub Actions** — Checks on commits
→ **07 Docker build** — Pinned dependencies
→ **08 Artifact Registry** — Versioned images
→ **09 Cloud Run API** — Candidate + smoke test
→ **10 Dashboard** — Forecast + model ID

Cloud Storage: versioned data + experiment artifacts. Promotion: checks + recorded approval.
Time order preserved / tune on validation / final test reported separately / no future AQI features

02 DEPLOYMENT STRATEGY

Docker + managed scaling

SCALE

Stateless FastAPI on Cloud Run. Scale to zero; service cap 3; tagged candidates add capacity.

RELEASE + ROLLBACK

Deploy candidate revision, smoke-test, then approve traffic switch. Restore prior revision if needed.

DEPLOYMENT CHALLENGES

Cold starts: small image. Model mismatch: pinned digest. Access: OIDC and least-privilege IAM.

03 MONITOR + RETRAIN

CO3 / feedback after actuals arrive

Forecast — Log model version
→ **Actual AQI** — Join by target date
→ **Error + drift** — MAE / input shift
→ **Review candidate** — Retrain + rerun gate

Cloud Logging: latency + failures. Drift prompts review; models never auto-promote.

04 GOVERNANCE + RESPONSIBLE AI

Poster criterion 2 / CO4

VERSIONING

Data SHA-256, Git SHA, MLflow run ID and container digest.

AUDIT TRAIL

Metrics, timestamp, release actor and deployed revision recorded.

COMPLIANCE

Public aggregate data; source attribution; IAM; log retention policy.

BIAS MITIGATION

Report seasonal errors; inspect extreme-AQI failures; disclose coverage gaps.

VERIFIED: GitHub CI/CD > private Cloud Run > smoke test > rollback + restore

2025 backtest MAE: forest 30.04 / persistence 29.68. Experimental forecasts; no accuracy guarantee.

Training: UrbanEmissions / CPCB. Live: official CPCB PDFs. Mumbai / OIDC / versioned artifacts.